

Parametric and Non-Parametric Methods

Exercise 1

In this exercise, you will use the Breast Cancer Wisconsin data set to build a classifier for breast tumor diagnosis based on features extracted from a digitized image of a fine needle aspirate of a breast mass (more information can be found [Here](#)). The code for the exercise in Matlab is `Classifier_NBC_KNN.m`.

- (a) Read through the file `Classifier_NBC_KNN.m` and fill in the missing parts.
- (b) Discuss the results and the confusion matrix for each classifier.
- (c) Experiment with the width parameter for the kernel density estimator and the number of neighbors for the k -nearest neighbor.

Exercise 2 (Extra)

Assume we have N independent draws $x_1, \dots, x_N \in \{0, 1\}$ from a Bernoulli distribution $p_X(x) = \theta^x (1 - \theta)^{1-x}$ with parameter $\theta \in [0, 1]$ representing the probability of $X = 1$.

- (a) What is the maximum likelihood estimator for θ ?
- (b) If we add a prior distribution for θ to be a Beta distribution $p_\theta(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$ with parameters $\alpha, \beta > 0$ and $B(\alpha, \beta)$ is a constant dependent on α and β . Find the maximum a posteriori estimate of θ .